

AURA — NEW V1 MASTER ENGINEERING OVERVIEW

1. PROJECT IDENTITY

Project: AURA — Academic Utility Robotic Assistant

Type:

Voice-first autonomous departmental assistant robot + mobile/cloud academic platform.

Primary environment:

- Small academic department
- Single floor
- Known indoor environment
- Corridors, laboratories, offices and attendance area
- Controlled movement environment

Primary users:

- Faculty
- Students
- Department administrators

Primary physical workload:

- Approximately 5–6 controlled movements/day
- Mainly attendance-related movement
- Faculty-requested navigation
- Department assistance

New V1 budget ceiling:

₹45,000 maximum

Preferred compute:
ROCK 5B+ 8GB

Alternative compute candidate:
Axon Vicharak, but only after compatibility validation.

Long-term direction:
Industry-grade indoor service / academic assistant platform.

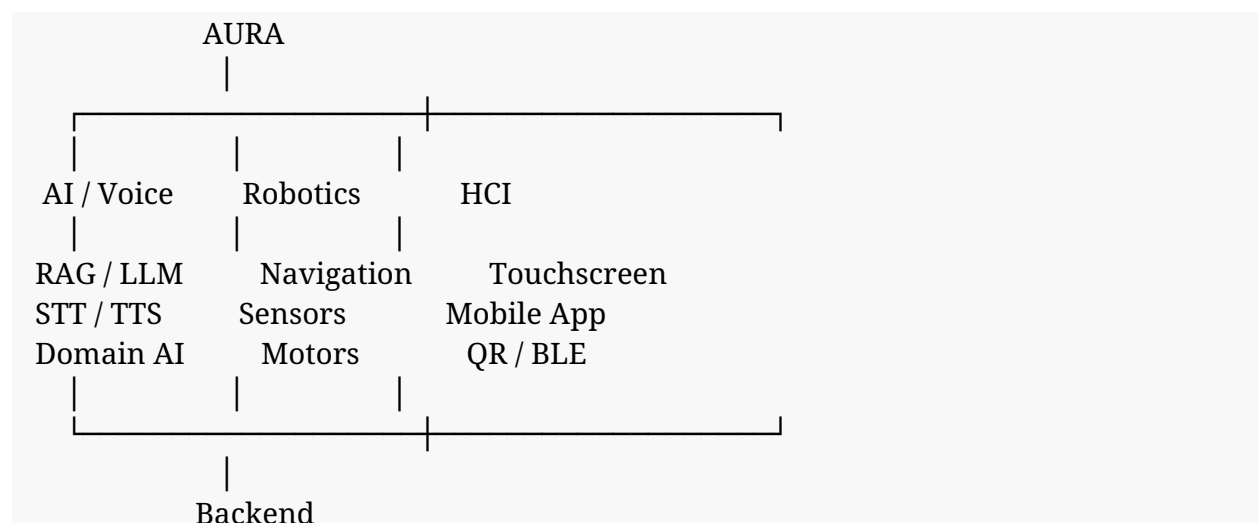
AURA is **not** intended to be:

- A household robot
- An unrestricted AI assistant
- An outdoor autonomous vehicle
- An industrial AGV in V1
- A security robot
- A medical robot

The project is deliberately constrained to **department-related operations and academic assistance**.

2. CORE PURPOSE OF AURA

AURA combines five major engineering areas:





The complete system should provide:

AI

- Voice interaction
- Department-specific Q&A
- RAG
- Local LLM
- Offline STT
- Offline TTS
- Multimodal educational assistance

Robotics

- Autonomous navigation
- Waypoint navigation
- LiDAR localization
- Encoder odometry
- IMU assistance
- Obstacle avoidance
- Closed-loop motor control

HCI

- Touchscreen
- Mobile application
- Voice interface
- QR display

- BLE attendance interaction

Department services

- Attendance
- Timetable
- Faculty information
- Notices
- Rooms/labs
- Syllabus
- Student/faculty operations

Backend

- Authentication
- RBAC
- Database
- AI routing
- RAG
- Robot APIs
- Logs
- Attendance APIs

This overall system structure follows the detailed project architecture in the uploaded overview.

3. THE BIGGEST ARCHITECTURAL CHANGE

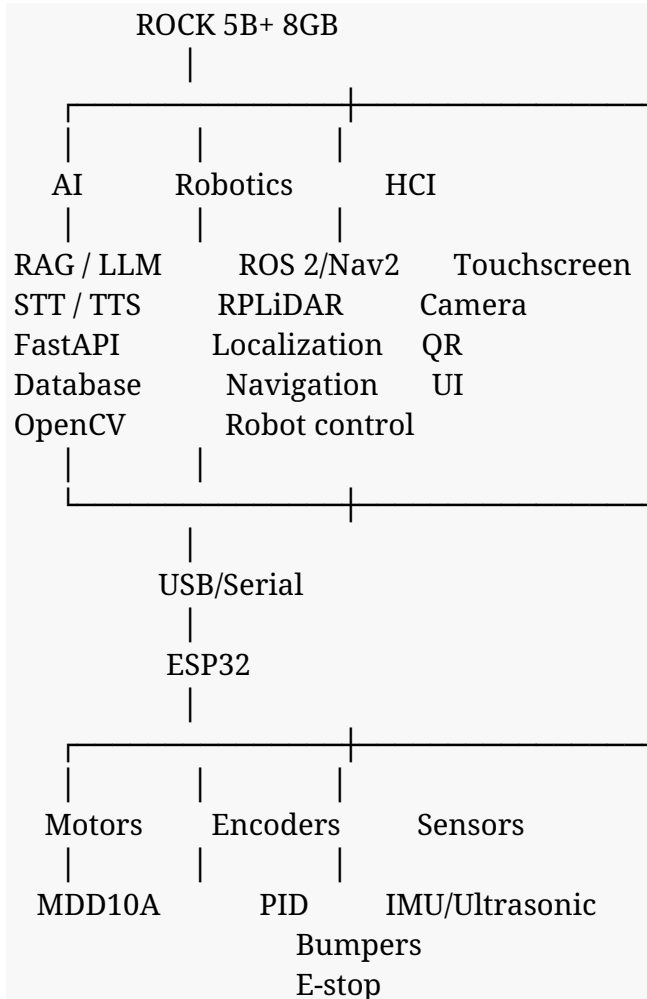
The old architecture was:

Raspberry Pi 4 8GB

+

Raspberry Pi 4 4GB
+
ESP32

The new V1 architecture is:



This is one of the most important design decisions.

The high-level computer is responsible for **intelligence and orchestration**.

The ESP32 is responsible for **deterministic real-time control and safety-related low-level behaviour**.

The uploaded architecture explicitly establishes this separation.

4. AURA'S CENTRAL ENGINEERING PRINCIPLE

ROCK decides what should happen.

For example:

User:

"AURA, take me to the HOD office."

↓

STT

↓

Intent:

NAVIGATE

↓

Destination:

HOD_OFFICE

↓

ROS 2 / Nav2

↓

Velocity command

↓

ESP32

ESP32 decides whether the low-level action can safely happen.

ESP32

Check:

└─ E-stop

|— bumper
|— watchdog
|— motor state
|— encoder state
|— sensor state
|— control limits

↓

Safe?

YES → Motor command

NO → STOP

This prevents a software failure from directly becoming an uncontrolled motor action.

5. COMPUTING PLATFORM

5.1 Preferred V1: ROCK 5B+ 8GB

AURA's preferred high-level computer is:

ROCK 5B+ — 8GB

The uploaded architecture identifies the ROCK 5B/5B+ family as the current computing platform and assigns it Linux, ROS 2, Nav2, RPLiDAR, FastAPI, database, RAG, local LLM, STT, TTS, OpenCV, camera, touchscreen and orchestration duties.

The RK3588 platform provides:

- 4 × Cortex-A76
- 4 × Cortex-A55
- Mali-G610 GPU
- NPU capability
- Hardware video acceleration

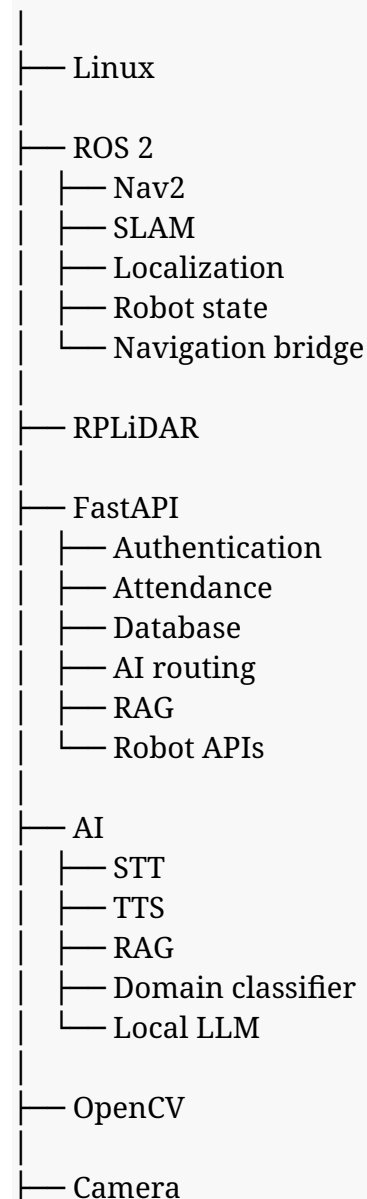
- High-speed storage
- Multiple display interfaces

The 5B+ additionally provides expanded interfaces and connectivity.

6. WHAT THE ROCK DOES

The ROCK is effectively AURA's **main brain**.

ROCK 5B+



- Touchscreen
- QR generation
- Robot orchestration

The ROCK should **not** be used for high-frequency motor PID control.

7. ALTERNATIVE COMPUTE — AXON VICHAAARAK

The new version can evaluate **Axon Vichaarak** as an alternative high-level compute platform.

However:

This is not specified in the uploaded EDURO documents.

Therefore it should be treated as an **engineering candidate**, not an officially documented architecture replacement.

Before replacing ROCK, test:

Operating system

- Linux availability
- ARM64 compatibility
- Driver availability

Robotics

- ROS 2 Jazzy
- Nav2
- Gazebo development compatibility
- USB serial communication

Sensors

- RPLiDAR
- USB camera
- microphone
- touchscreen

AI

- sentence-transformers
- ChromaDB
- local LLM runtime
- STT
- TTS
- OpenCV

Performance

- CPU utilization
- RAM consumption
- sustained AI inference
- simultaneous ROS + AI workload
- thermal throttling

Power

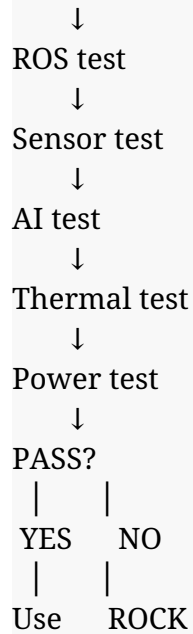
- operating voltage
- peak current
- cooling requirement

The replacement decision should therefore be:

Axon Vicharak

↓

Compatibility test



8. ESP32 REAL-TIME CONTROLLER

ESP32 remains essential.

Its responsibilities are:

ESP32

- |
- |— PWM
- |— Motor direction
- |— Encoder counting
- |— PID
- |— IMU
- |— Ultrasonic sensors
- |— Bumpers
- |— E-stop status
- |— Watchdog
- |— BLE beacon

This is directly consistent with the uploaded architecture.

Why not directly control motors from ROCK?

Because Linux is not a deterministic real-time motor-control environment.

For example:

```
Nav2
↓
10 Hz / velocity command
↓
ESP32
↓
high-frequency control loop
↓
Encoder feedback
↓
PID
↓
PWM
```

The ROCK tells the ESP32 the desired motion.

The ESP32 continuously corrects the actual motor behaviour.

9. COMPLETE MOTOR CONTROL LOOP

AURA uses:

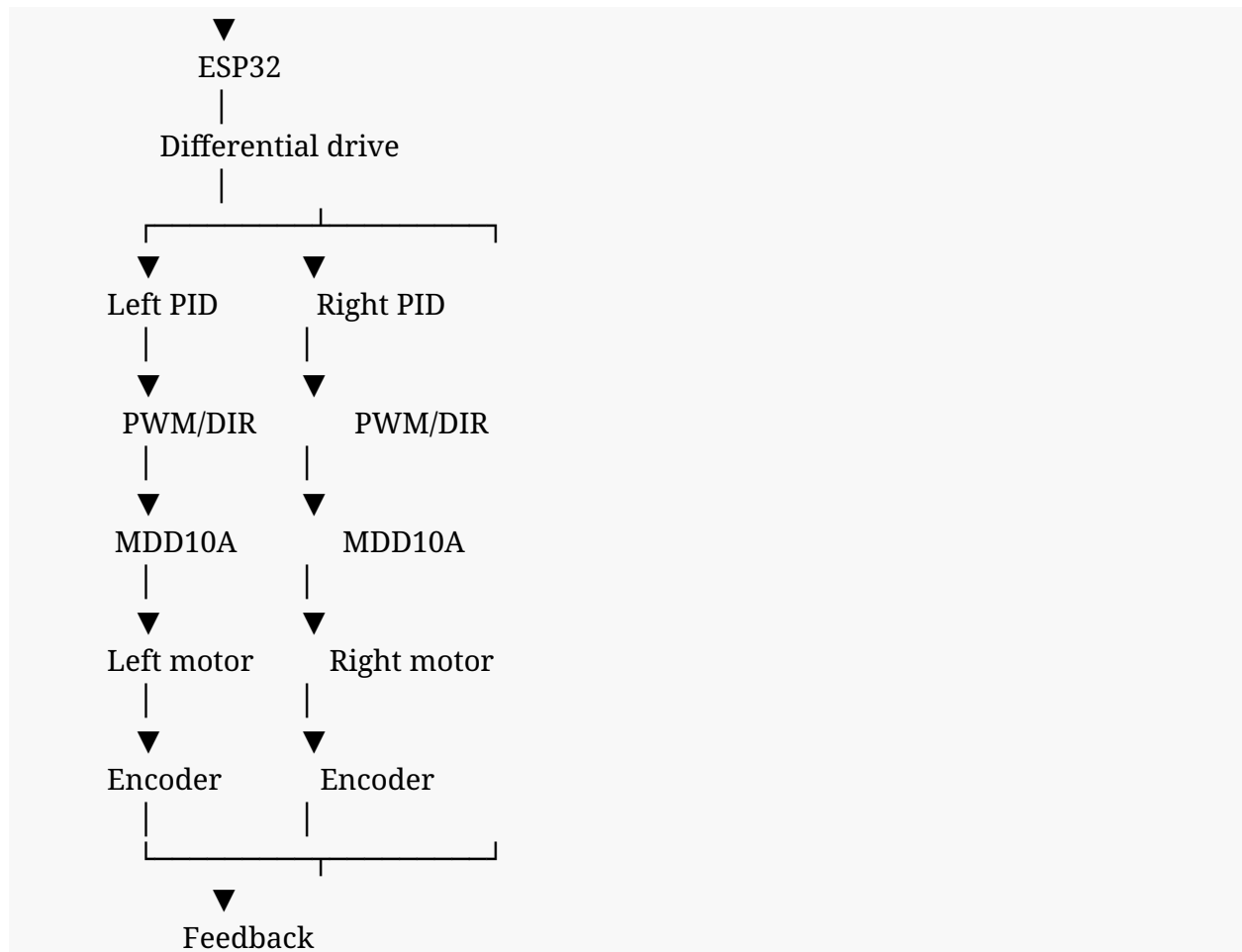
2 × 12V encoder DC geared motors

Target:

- Approximately 100 RPM
- Quadrature encoders
- Suitable torque for small indoor robot

Architecture:

```
Desired velocity
|
▼
ROS 2 / Nav2
|
/cmd_vel
|
```



This converts the robot from:

"Apply PWM and hope it goes straight."

to:

Measure → compare → correct → measure again.

10. MOTOR CHARACTERIZATION

Because the V1 uses cost-controlled motors, don't assume both motors behave identically.

Each motor should be characterized separately.

Measure:

Startup PWM

PWM = 10% → no movement

PWM = 15% → no movement

PWM = 20% → movement begins

This gives the motor's approximate dead zone.

PWM vs RPM

Measure:

PWM

↓

RPM

for several PWM levels.

Loaded RPM

Test the motor under the actual robot load.

Encoder counts

Determine:

encoder counts

↓

wheel rotation

↓

distance travelled

Left/right mismatch

For example:

Left = 92 RPM

Right = 98 RPM

Without feedback, the robot turns.

With PID:

Left PID → increase drive

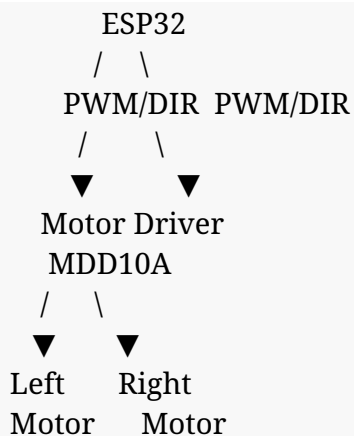
Right PID → reduce drive

11. MOTOR DRIVER

Current V1:

Cytron MDD10A-class dual motor driver

Architecture:



The motor driver is entirely below the ROS/Nav2 layer.

Before operation, verify:

- Motor voltage compatibility
 - Driver current capability
 - Motor stall current
 - Wiring gauge
 - Connector current rating
 - Driver thermal behaviour
 - Grounding
 - Fuse protection
-

12. IMU — MPU6050

The MPU6050 provides:

- Accelerometer
- Gyroscope
- Motion information
- Yaw-related information
- Motion diagnostics

Combined with encoders:

```
Encoders
+
MPU6050
↓
Motion estimation
↓
Odometry / orientation
↓
ROS 2 localization
```

The IMU should be:

- Rigidly mounted
- Centrally located where practical
- Axis-aligned with the robot frame
- Properly configured in ROS TF

Incorrect IMU orientation can create seemingly mysterious navigation errors.

13. PRIMARY NAVIGATION SENSOR — RPLIDAR A1M8

RPLiDAR is the primary navigation sensor.

Functions:

- 2D scanning

- Mapping
- Localization
- Obstacle detection
- Nav2 integration

Pipeline:

RPLiDAR



/scan



SLAM / Localization



Nav2



/cmd_vel



ROCK



ESP32



PID



Motors

The uploaded overview specifically establishes this RPLiDAR → ROS 2/Nav2 navigation pipeline.

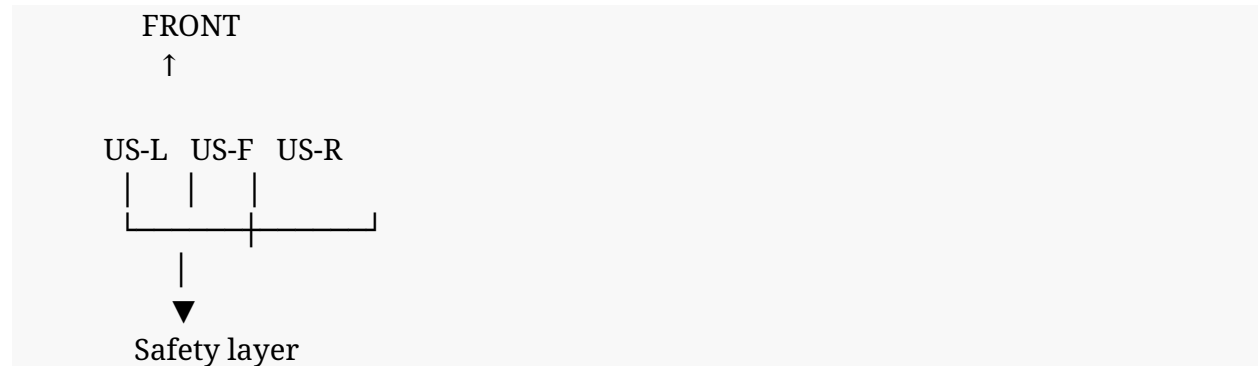
14. ULTRASONIC SAFETY LAYER

The new V1 deliberately retains:

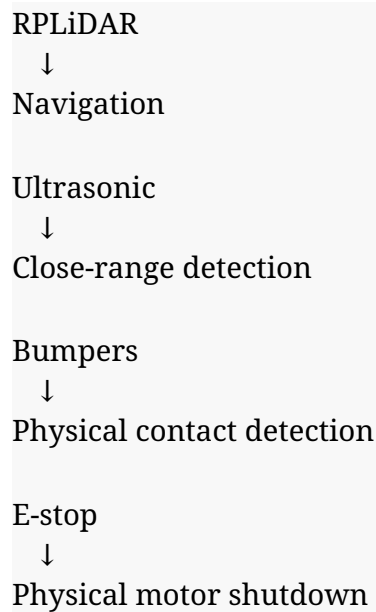
3 × HC-SR04

They are **not the primary mapping system**.

Their job is secondary close-range protection.



The complete layered safety model is:



This layered approach is appropriate for a slow indoor robot.

15. BUMPER SYSTEM

Use:

2 or more mechanical bumper/limit switches.

Behaviour:

Physical collision



Bumper switch



ESP32



Immediate STOP

Where practical, the bumper input should be designed so that wiring failure does not falsely appear as a healthy safety condition.

16. EMERGENCY STOP

This is non-negotiable.

Correct architecture:

Motor Battery



Fuse



PHYSICAL E-STOP



Motor Driver



Motors

The E-stop must **physically interrupt motor power**.

It must not depend on:

- Linux
- ROS 2
- Python

- Wi-Fi
- FastAPI
- LLM
- Cloud AI

For example:

Cloud server crashes

↓

No effect on E-stop

ROCK crashes

↓

E-stop still works

Wi-Fi disconnects

↓

E-stop still works

This physical safety architecture is explicitly present in the source overview.

17. CAMERA

Current V1:

USB UVC camera

Functions:

- QR scanning
- OpenCV
- Faculty recognition
- Visual diagnostics
- Future computer vision

Pipeline:

USB Camera



OpenCV



Detection



Recognition / QR / CV

The V1 deliberately avoids making the camera subsystem unnecessarily complex.

18. CONTROLLED FACULTY RECOGNITION

AURA's recognition system should be treated as a **closed registered-faculty recognition system**.

Not:

"Identify anyone in the building."

Instead:

Camera



Face detection



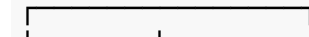
Face embedding



Compare against registered faculty



Known?



YES



Faculty
match



Personalized UI

NO



Generic
interaction

The source architecture previously described approximately 6–7 registered faculty members.

Example

Camera detects person

↓

Face embedding

↓

Registered faculty database

↓

Match: Faculty A

↓

"A good morning, Professor."

↓

Quick Actions

Recognition should not be allowed to become a single point of failure for the voice interface.

19. TOUCHSCREEN

Current V1:

7-inch HDMI touchscreen

Main UI:

HOME

|

|— Faculty

- Attendance
- Timetable
- Notices
- Faculty Information
- Robot Status
- Navigation
- QR

The touchscreen has two roles:

Primary HCI

Users can interact directly.

Fallback HCI

If:

- microphone fails
- STT fails
- environment is noisy
- user doesn't want voice

the touchscreen remains available.

20. AUDIO SYSTEM — NEW V1

The new V1 should use a **budget but functional audio system**.

Current architecture:

USB Microphone

↓

VAD

↓

STT

↓

AI / Intent

↓

TTS

↓
Amplifier
↓
Speaker

Hardware:

- USB microphone
- Small/Class-D amplifier
- Speaker

The original architecture also identifies a microphone array as a future upgrade rather than an essential low-budget V1 component.

V1 objective

Not:

perfect far-field speech in a noisy corridor.

Instead:

reliable conversational voice interaction at a reasonable distance in a reasonably controlled department environment.

21. AUDIO ENGINEERING

Important physical considerations:

Microphone placement

Do not mount:

microphone
↓
directly beside
↓
speaker

because this increases acoustic feedback.

Instead:

MIC ----- SPEAKER

with physical separation where possible.

Motor noise

Keep microphone away from:

- motor driver
- motors
- cooling fans
- high-current wiring

Software

Use:

- VAD
 - noise handling
 - audio normalization
 - controlled TTS volume
-

22. POWER ARCHITECTURE

AURA uses two electrical domains.

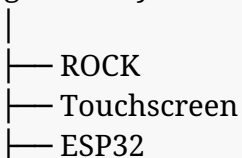
Logic battery

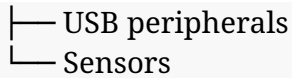
Target:

3S 11.1V ~5200mAh

Powers:

Logic battery





through appropriate regulators.

Motor battery

Target:

3S 11.1V ~2200mAh high-discharge

Powers:

Motor battery



Fuse



E-stop



Motor driver



Motors

This dual-domain architecture is explicitly defined in the uploaded overview.

23. WHY TWO BATTERIES?

Motors generate:

- Current spikes
- Electrical noise
- Voltage disturbances
- Regenerative effects

Bad architecture:

Motor current spike



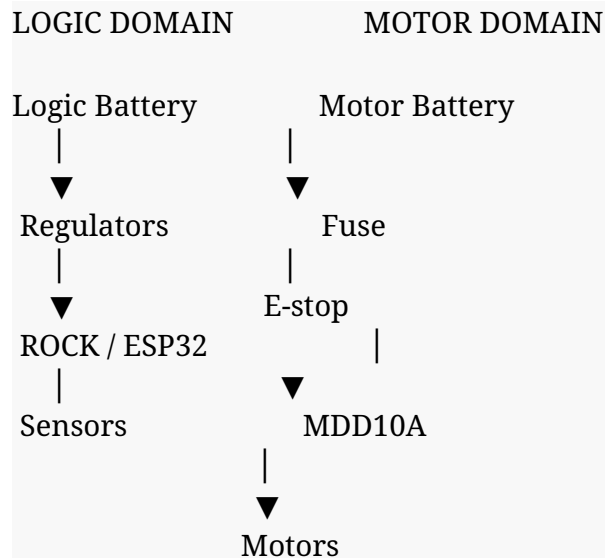
Logic voltage disturbance



ROCK crash

↓
Robot failure

Preferred architecture:



However, signal grounding must still be engineered correctly.

Separate batteries do **not** automatically solve every grounding/noise problem.

24. MECHANICAL DESIGN

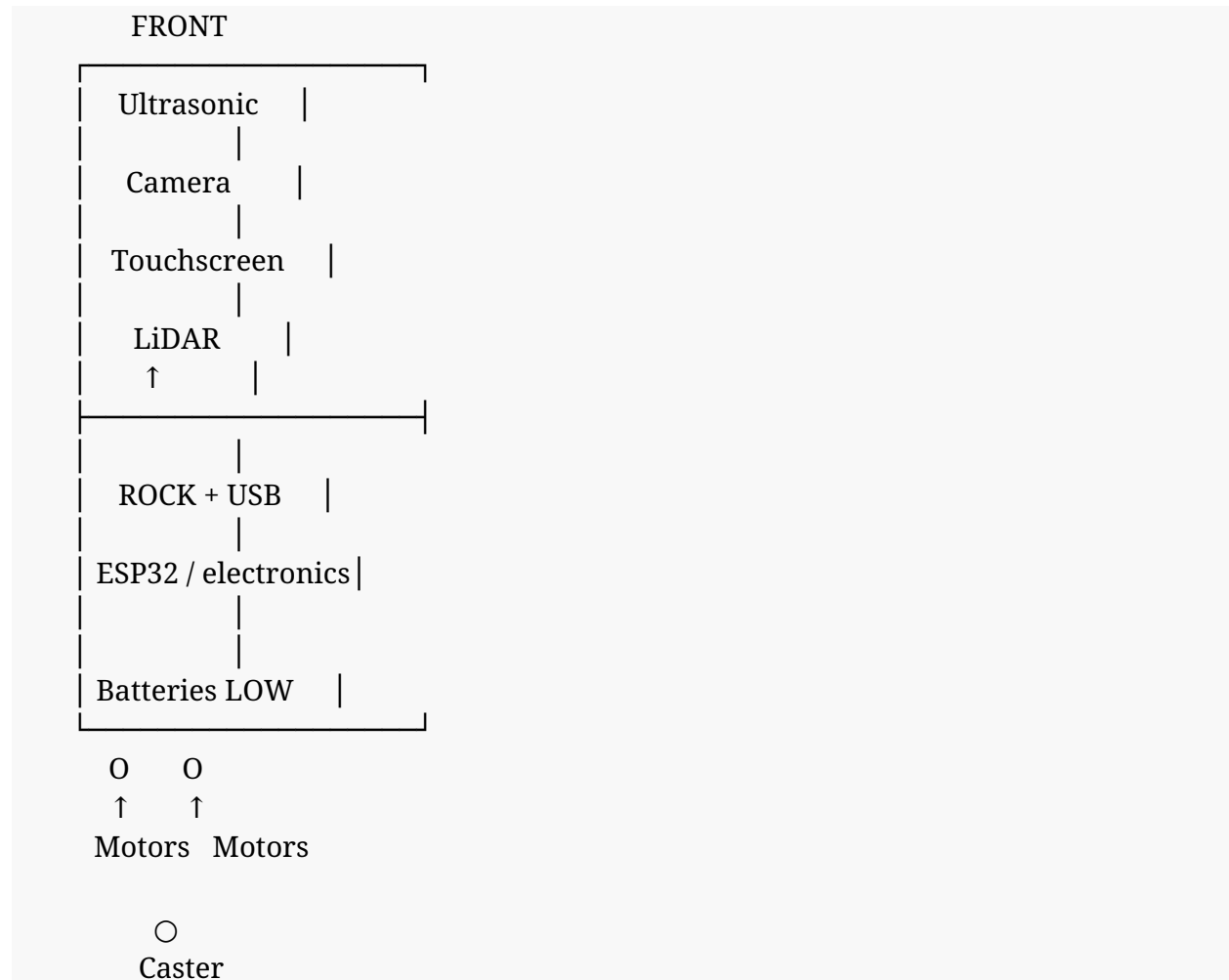
Recommended V1 target:

- Width: ~400–450 mm
- Depth: ~350–400 mm
- Wheel diameter: ~90–100 mm
- 2 powered wheels
- 1 caster
- Low battery placement
- Elevated LiDAR
- User-friendly display height

The source overview specifically recommends these approximate dimensions and states that final dimensions should be frozen during digital-twin development.

25. MECHANICAL LAYOUT

Conceptually:



The exact mechanical dimensions should not be finalized before simulation.

26. NAVIGATION PHILOSOPHY

AURA does **not** need to solve arbitrary autonomous navigation.

The environment is:

- Small
- Indoor
- Single floor
- Known
- Controlled
- Low traffic compared with public robots
- Approximately 5–6 trips/day
- Mostly fixed destinations

Therefore:

Known Map

+

Named Waypoints

+

LiDAR Localization

+

Encoder Odometry

+

IMU

+

Obstacle Avoidance

+

Optional ArUco

↓

Predictable Navigation

This is much more appropriate than designing an unnecessarily fast autonomous robot.

27. WAYPOINT SYSTEM

Define named locations.

For example:

```
HOME
ATTENDANCE_POINT
HOD_OFFICE
ELECTRONICS_LAB
COMPUTER_LAB
```

Then:

```
User command
↓
Destination name
↓
Waypoint lookup
↓
Nav2
↓
Navigation
```

This is much easier to control than asking an AI model to generate arbitrary coordinates.

28. ATTENDANCE NAVIGATION

AURA does not need to enter every classroom.

Actual workflow:

```
AURA
↓
Attendance Point
↓
STOP
↓
Attendance Session
↓
Generate QR
↓
Display QR
↓
Students scan
```

Smartboards can also display the attendance interface.

This substantially reduces:

- navigation complexity
 - collision risk
 - classroom disruption
 - battery consumption
 - localization requirements
-

29. ARUCO LANDMARKS

Optional low-cost navigation improvement:

ArUco markers

Possible locations:

HOME
ATTENDANCE_POINT
HOD_OFFICE
ELECTRONICS_LAB
COMPUTER_LAB

Architecture:

LiDAR
+
Encoder
+
IMU
+
ArUco
↓
Localization

ArUco is not replacing LiDAR.

It provides an additional visual landmark for:

- recovery
 - precise stopping
 - known-location verification
-

30. ROBOT SPEED

Starting range:

~0.15–0.25 m/s

Priority:

Safety

>

Reliability

>

Repeatability

>

Low latency

>

Speed

The robot can remain parked most of the day and perform controlled trips.

There is therefore little engineering benefit in maximizing speed in V1.

31. ROS 2 + GAZEBO STACK

Software simulation environment:

Ubuntu 24.04

+

ROS 2 Jazzy

+

Gazebo Harmonic

+

ros_gz_bridge

ROS 2 handles:

- Robot communication
- Navigation
- Sensor topics
- TF
- Odometry
- Navigation commands

Gazebo provides:

- Robot simulation
 - Environment
 - Sensors
 - Physics
 - Obstacle scenarios
-

32. DIGITAL TWIN

Before physically assembling the final robot:

AURA

↓

URDF

↓

Gazebo

↓

Department environment

↓

Sensor simulation

↓

Odometry

↓

LiDAR

↓

Nav2

↓

Obstacle testing

The Month-1 requirement specifically includes:

- Robot dimensions
- Wheel radius
- Wheel separation
- Sensor positions
- Coordinate frames
- ROS topics

as part of the digital-twin specification.

33. WHAT TO TEST IN THE DIGITAL TWIN

Mechanical parameters

- Wheel radius
- Track width
- Chassis dimensions

Sensors

- LiDAR position
- Camera position
- IMU position
- Ultrasonic positions

Navigation

- Straight-line motion
- Turning

- Waypoint navigation
- Obstacle avoidance
- Goal stopping

ROS

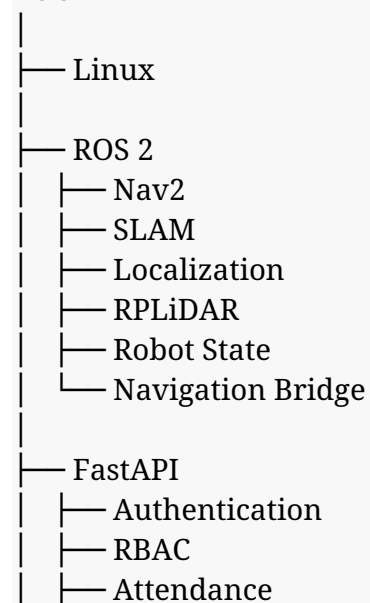
- /cmd_vel
- /odom
- /scan
- TF
- Robot state

Safety

- Simulated bumper event
 - Emergency stop state
 - Obstacle stopping
-

34. SOFTWARE ARCHITECTURE

ROCK

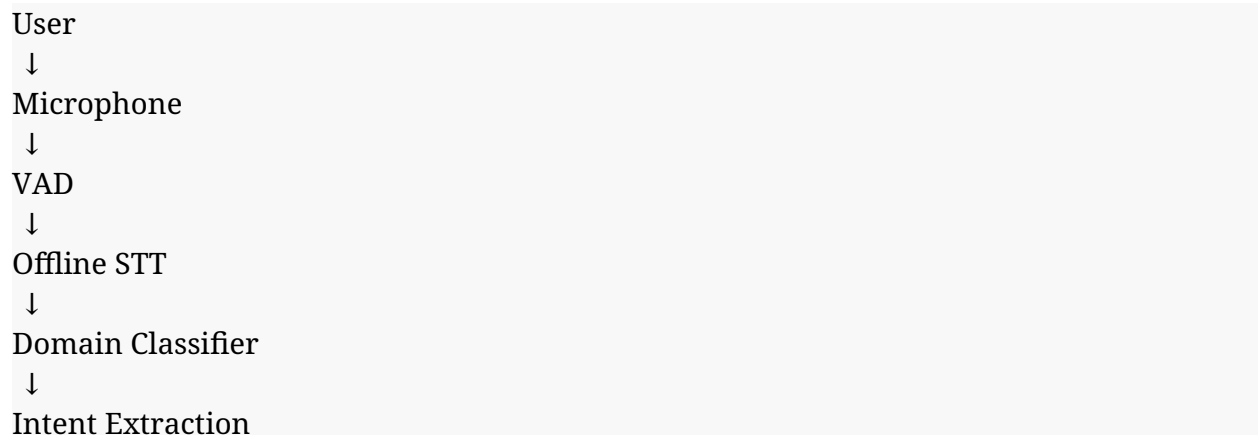


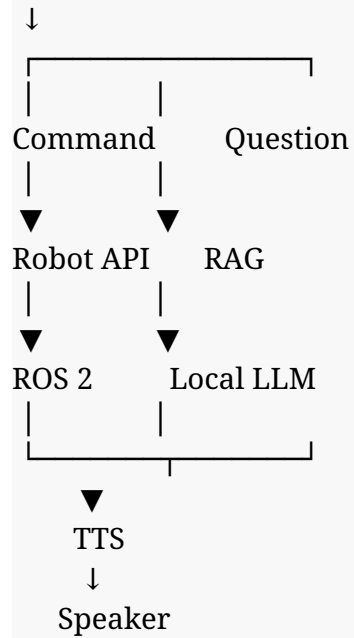


This matches the uploaded software architecture.

35. VOICE PIPELINE

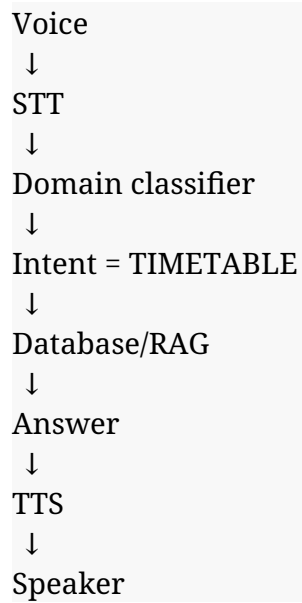
Complete voice pipeline:





Example:

"What is the Digital Electronics timetable?"



36. LOCAL AI PHILOSOPHY

The local LLM is **not the source of truth**.

The source of truth is:

Department Data
↓
Database / RAG
↓
Verified Context
↓
LLM
↓
Natural Language

Not:

User
↓
LLM
↓
Whatever the model thinks

This architecture reduces hallucination.

37. DOMAIN CLASSIFIER

Before calling the LLM:

User Query
↓
Domain Classifier
├── Department-related
│ ↓
│ RAG
│ ↓
│ LLM
└── Out of scope
 ↓
 Refusal

Example:

Allowed

"Explain transformers."

→ Academic topic → allowed.

"Where is the Electronics Lab?"

→ Department operation → allowed.

"What is today's timetable?"

→ Department information → allowed.

Rejected

"Who won yesterday's cricket match?"

→ Outside the defined department scope.

The uploaded overview explicitly establishes this domain-controlled interaction model.

38. DEPARTMENT KNOWLEDGE BASE

Planned knowledge base:

27 Electrical Engineering subjects

Across:

- Semester 3
- Semester 4
- Semester 5
- Semester 6
- Semester 7
- Semester 8

With:

6 units per subject

Knowledge categories:

Syllabus
Faculty
Timetable
Rooms
Labs
Notices
FAQs
Routes
Attendance
Department Policies
Operational Information

39. RAG PIPELINE

Department Documents

↓

Cleaning

↓

Chunking

↓

Metadata

↓

Embeddings

↓

ChromaDB

↓

Semantic Retrieval

↓

Context Validation

↓

LLM

↓

Answer

Example metadata:

JSON

```
{  
  "subject": "Digital Electronics",  
  "semester": 4,  
  "unit": 3,
```

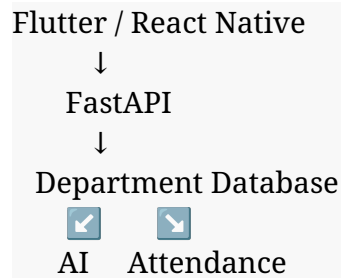


```
"source": "official_syllabus"
}
```

This allows retrieval to be constrained to relevant material.

40. MOBILE APPLICATION

Mobile is a separate but connected AURA subsystem.



The mobile application supports both students and faculty.

41. STUDENT FREE TIER

Functions defined in the source architecture include:

- Attendance
- Attendance analytics
- Syllabus AI
- Faculty/operational Q&A
- Notices
- Mentor issues
- Project archives

Proposed AI provider fallback:

```
Groq
↓
```

Gemini

↓

OpenRouter

This is an application-level fallback.

It must not be placed inside the robot's safety or motor-control path.

42. PAID STUDENT TIER

Proposed functions:

- Priority AI routing
- Gemini multimodal processing
- PDF analysis
- Handwritten-note analysis
- Circuit/schematic analysis
- CRO/DSO waveform analysis
- Practical guidance
- Attendance prediction
- Mentor escalation

The original proposal placed the paid tier around ₹150–₹200/year, but actual provider economics need to be revalidated when deployment begins.

43. CLOUD AI SAFETY BOUNDARY

Absolutely critical:

Wrong

Cloud AI

↓

"Move forward"

↓
Motor

Correct

Cloud AI
↓
High-level intent
↓
ROS 2
↓
Nav2
↓
Velocity command
↓
ESP32
↓
Safety checks
↓
PID
↓
Motor

If the cloud disappears:

Internet OFF
↓
Cloud AI unavailable
↓
Physical motor safety unaffected

44. ATTENDANCE ARCHITECTURE

AURA attendance:

Student Mobile App
|
├─ Device Binding
├─ BLE Proximity
└─ TOTP QR
↓
FastAPI



Attendance Database

Three mechanisms work together:

1. Device binding
 2. BLE proximity
 3. Dynamic TOTP QR
-

45. DEVICE BINDING

The student's account is associated with a device identifier.

Purpose:

Student account



Registered device



Reduced account sharing

The exact identifier implementation must respect mobile-platform privacy/security constraints.

46. BLE PROXIMITY

ESP32 advertises:

AURA_ATTENDANCE_BEACON

Phone:

Detect BLE



Check proximity condition



Allow attendance workflow

The source architecture proposes approximately **3 m**.

However, BLE RSSI is not a precise distance measurement.

Therefore:

3 m
↓
Initial engineering assumption
↓
Real-world calibration
↓
Final threshold

47. DYNAMIC TOTP QR

Workflow:

TOTP Secret
↓
Current Time
↓
OTP
↓
QR
↓
Touchscreen
↓
Student Scan
↓
Backend Validation

The QR refreshes every **10 seconds** according to the current project specification in the uploaded overview.

The backend must validate the code.

Never trust the client-side QR result directly.

48. ACTUAL ATTENDANCE DEPLOYMENT

AURA does **not** need to drive inside classrooms.

Instead:

```
HOME
↓
ATTENDANCE POINT
↓
STOP
↓
Attendance session
↓
QR displayed
↓
Students scan
↓
Session closes
↓
HOME
```

This significantly simplifies:

- Navigation
 - Safety
 - Classroom interaction
 - Localization
 - Battery usage
-

49. FACULTY NAVIGATION WORKFLOW

Example:

"AURA, take me to the HOD office."

Pipeline:

```
Voice
↓
STT
↓
Domain classifier
```

↓
Intent = NAVIGATE
↓
Destination = HOD_OFFICE
↓
Nav2
↓
/cmd_vel
↓
ESP32
↓
PID
↓
Motors
↓
Navigation complete
↓
" We have reached the HOD office."

This is a complete cross-domain integration test.

50. FACULTY MOBILE FUNCTIONS

Faculty interface can include:

- Lecture completion
- Syllabus coverage
- Attendance
- Attendance analytics
- Notices
- Student issues
- Infrastructure tickets
- Excel reports

These functions are part of the uploaded architecture.

51. AURA DASHBOARD

Dashboard:

AURA Dashboard

- |
- |— Robot Status
- |— Battery
- |— Location
- |— Navigation
- |— Attendance
- |— Timetable
- |— Faculty
- |— Notices
- |— AI
- |— Issues
- |— Logs

Robot controls must be authenticated and permission-controlled.

52. BACKEND

Primary backend:

FastAPI

Responsibilities:

FastAPI

- |
- |— Authentication
- |— RBAC
- |— Attendance
- |— QR
- |— BLE state
- |— Robot status
- |— Navigation requests
- |— AI routing
- |— RAG access

- Database
- Notices
- Faculty
- Timetable
- Logs

53. DATABASE

Initial development:

SQLite

Expanded/production deployment:

PostgreSQL

Core entities:

- Students
- Faculty
- Subjects
- Units
- Rooms
- Timetable
- Notices
- Attendance_Logs
- Mentor_Issues
- Robot_Status
- Navigation_Points
- AI_Logs

This database model is directly supported by the uploaded overview.

54. ROLE-BASED ACCESS CONTROL

Minimum roles:

- Faculty
- Free_Student

Paid_Student
Admin

Student

Can access:

- Attendance
- Q&A
- Notices
- Syllabus
- Personal learning tools according to tier

Faculty

Can access:

- Attendance analytics
- Notices
- Syllabus management
- Issue management
- Robot interaction

Admin

Can access:

- System configuration
 - Users
 - Knowledge base
 - Robot management
 - Logs
 - Security settings
-

55. FIVE ENGINEERING DOMAINS

AURA is divided into five major engineering domains.

Domain 1 — Electronics & Electrical

Responsible for:

- Power
 - Motor driver
 - PCB
 - E-stop
 - Sensors
 - Electrical safety
 - Battery management
 - Power distribution
-

Domain 2 — Computer Science / Software

Responsible for:

- Linux
- FastAPI
- Database
- APIs
- Authentication
- AI routing
- Attendance
- Mobile backend
- Logging

Domain 3 — AI & Computer Vision

Responsible for:

- STT
 - TTS
 - RAG
 - Embeddings
 - Domain classifier
 - Local LLM
 - OpenCV
 - Faculty recognition
 - Multimodal AI
-

Domain 4 — Robotics & Mechanical

Responsible for:

- Chassis
- Motors
- Encoders
- PID
- Odometry
- LiDAR
- SLAM
- Nav2
- Obstacle avoidance

- Simulation
-

Domain 5 — IoT / Networking / HCI

Responsible for:

- Wi-Fi
- BLE
- Mobile application
- Touchscreen
- WebSockets
- Device binding
- Attendance
- UI/UX
- Robot/cloud communication

The five-domain responsibility structure is explicitly defined in the uploaded domain architecture.

56. COMPLETE SOFTWARE STACK

Operating system

Ubuntu 24.04 / Linux

Robotics

ROS 2 Jazzy

Nav2

SLAM / Localization

Gazebo Harmonic

ros_gz_bridge

Backend

Python
FastAPI
SQLite
PostgreSQL

AI

sentence-transformers
ChromaDB
Local LLM
Offline STT
Offline TTS

Computer vision

OpenCV
MediaPipe / lightweight recognition

Mobile

Flutter
OR
React Native

Development

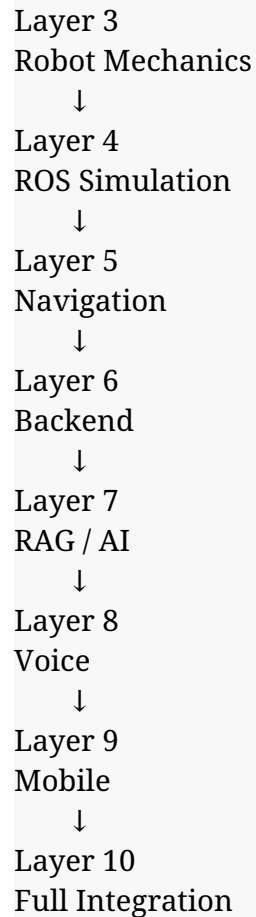
Git / GitHub
Docker
WSL2
Ubuntu 24.04

57. DEVELOPMENT ORDER

Do **not** build all components simultaneously.

Correct order:

Layer 1
Electronics
↓
Layer 2
ESP32
↓



```
graph TD; L3[Layer 3  
Robot Mechanics] --> L4[Layer 4  
ROS Simulation]; L4 --> L5[Layer 5  
Navigation]; L5 --> L6[Layer 6  
Backend]; L6 --> L7[Layer 7  
RAG / AI]; L7 --> L8[Layer 8  
Voice]; L8 --> L9[Layer 9  
Mobile]; L9 --> L10[Layer 10  
Full Integration];
```

Layer 3
Robot Mechanics
↓
Layer 4
ROS Simulation
↓
Layer 5
Navigation
↓
Layer 6
Backend
↓
Layer 7
RAG / AI
↓
Layer 8
Voice
↓
Layer 9
Mobile
↓
Layer 10
Full Integration

This layered development strategy is explicitly specified in the source architecture.

58. MONTH 1

Month 1 is:

Foundation + Engineering Setup

It is **not final robot assembly**.

Required areas:

- Team fundamentals
- Development environment
- Git

- Electronics simulation
 - ESP32 simulation
 - ROS 2
 - Gazebo
 - FastAPI
 - AI/RAG experiment
 - Basic mobile/UI prototype
-

59. MONTH 1 FINAL DIGITAL-TWIN DEMO

The expected software-first foundation is:

2WD AURA

+

URDF

+

Gazebo

+

cmd_vel

+

odom

+

scan

+

camera

+

FastAPI

+

dashboard

+

basic RAG

+

domain classifier

The important point is:

Don't wait for the physical robot to begin integration.

The architecture should be validated in simulation first.

60. NEW V1 ₹45K ENGINEERING PHILOSOPHY

The new V1 has a hard target of:

₹45,000

Therefore the design philosophy becomes:

Premium hardware everywhere

↓

Too costly

X

Cost-controlled hardware

↓

Characterization

↓

Calibration

↓

Software compensation

↓

Testing

↓

Reliable prototype

The key is **not** to remove safety-critical architecture.

61. WHAT MUST NOT BE SACRIFICED

Even under ₹45K, protect:

Safety

- Physical E-stop
- Fuse

- Bumpers
- Ultrasonic layer
- Watchdog

Navigation

- RPLiDAR
- Encoders
- IMU

Motion control

- Closed-loop PID
- Proper motor driver

Electrical architecture

- Dual power domains
- Correct regulation
- Correct grounding

Compute

- Adequate high-level compute
-

62. WHERE THE BUDGET CAN BE REDUCED

Cost can be controlled through:

- Commodity chassis
- Budget display
- Standard USB camera
- Standard USB microphone
- Budget speaker/amplifier

- Cost-efficient encoder motors
- Modular prototype electronics
- Local fabrication

But the following should **not** be removed merely to save money:

E-stop
Fuse
Encoder feedback
LiDAR
Motor protection
Power separation
Watchdog

63. ₹45K V1 FUNCTIONAL ARCHITECTURE

The complete budget-oriented V1 still contains:

COMPUTE

ROCK 5B+ / 5B 8GB
NVMe

CONTROL

ESP32

DRIVE

2 × encoder motors
MDD10A-class driver

NAVIGATION

RPLiDAR
MPU6050
3 × HC-SR04

SAFETY

Bumpers
E-stop
Fuse
Watchdog

VISION

USB camera

HCI

7" touchscreen

AUDIO

USB mic

Amplifier

Speaker

POWER

Logic battery

Motor battery

DC/DC regulators

MECHANICS

2WD

2 wheels

Caster

Mounts

SOFTWARE

ROS 2

Nav2

Gazebo

FastAPI

RAG

AI

Mobile

The source BOM contains this same overall subsystem structure.

64. EXPECTED V1 COMPROMISES

The ₹45K version will have limitations.

Motors

Possible:

- Greater motor-to-motor mismatch
- Gearbox backlash
- Lower consistency

Solution:

Characterization

+

Encoder

+

PID

+

Calibration

Chassis

Possible:

- Slight dimensional error
- More vibration
- Less rigidity

Solution:

- Accurate fabrication
- Proper mounting
- Geometry calibration

Camera

Possible:

- Lower low-light performance
- Lower dynamic range

Solution:

- Controlled lighting
- Fixed camera position
- Small registered faculty set

Audio

Possible:

- Lower far-field performance
- More sensitivity to environmental noise

Solution:

- Reasonable speaking distance
 - VAD
 - Microphone positioning
 - Speaker isolation
-

65. MOTOR + ODOMETRY CALIBRATION

Do not immediately blame Nav2 when navigation is wrong.

Debug in this order:

Mechanical alignment



Wheel dimensions



Encoder calibration



Motor PID



Odometry



IMU



LiDAR localization

↓
Nav2

For example, if AURA constantly curves left:

Possible causes:

Left motor weaker

OR

Right motor stronger

OR

Wheel diameters differ

OR

Track width incorrect

OR

Encoder scale wrong

OR

PID poorly tuned

Only after those are verified should Nav2 parameters be changed.

66. DIFFERENTIAL DRIVE CALIBRATION

Measure:

Wheel diameter

Actual, not nominal.

Wheel separation

Actual distance between wheel contact centres.

Encoder scale

Counts / revolution

Straight-line error

Command:

1 metre forward

Measure:

Actual:
0.96 m

Then calibrate.

Rotation error

Command:

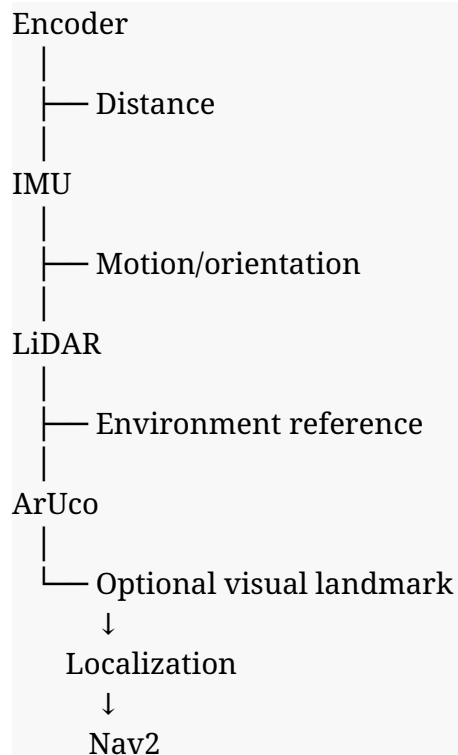
360°

Measure actual angle.

Repeat until the error becomes repeatable.

67. LOCALIZATION ARCHITECTURE

AURA should not rely on one sensor.



This is especially useful with cost-controlled motors.

68. AI RESPONSE LATENCY

The new V1 should prioritize deterministic responses.

For example:

"Where is the Electronics Lab?"

There is no reason to perform a large LLM operation if the database already knows:

Electronics Lab

→ First Floor

→ Room X

Instead:

Voice

↓

STT

↓

Intent

↓

Database

↓

Answer

↓

TTS

This can be much faster.

For complex questions:

Voice

↓

STT

↓

RAG

↓

Local LLM

↓

TTS

69. FACULTY RECOGNITION + VOICE COMBINATION

A strong interaction is:

Camera



Recognize registered faculty



Faculty profile loaded



Voice command



Intent



Personalized action

Example:

Recognized:

Faculty A

Voice:

"Take me to the HOD office."



Faculty profile

+

Voice intent



Navigation

Recognition therefore becomes a personalization mechanism rather than the robot's entire identity system.

70. INDUSTRY-GRADE UPGRADE PATH

The new V1 should be **upgrade-ready**.

The evolution should be:

AURA V1
₹45K prototype
↓
Reliable department robot
↓
AURA V2
Better electronics + mechanics + audio
↓
AURA V3
Advanced AI + vision + navigation
↓
Industrial AURA
↓
Fleet + Analytics + OTA + Security

The source architecture similarly describes the evolution from a department prototype to an industry-grade multi-robot platform.

71. FUTURE COMPUTE

Current:

ROCK 5B+ 8GB

Future:

- Higher-RAM ROCK platform
- Industrial RK3588 platform
- Dedicated AI accelerator where justified
- Industrial ARM compute
- Potential Jetson-class platform if advanced vision/AI requires it

Future RAM:

16 GB
24 GB
32 GB

depending on:

- Local model size
- Multiple CV pipelines
- Vector database size
- Concurrent services

The source overview identifies higher-RAM and industrial compute as future upgrade directions.

72. FUTURE AUDIO

Current:

USB microphone
+
amplifier
+
speaker

Future:

4–8 microphone array
↓
Beamforming
↓
Noise rejection
↓
Echo cancellation
↓
Better far-field speech

The original architecture specifically identifies microphone arrays as a future upgrade.

73. FUTURE MOTORS

Current:

12V encoder DC motors

Future:

Higher-quality geared motors

↓

BLDC

↓

Servo / industrial drive

Goals:

- Better efficiency
 - Lower backlash
 - Better reliability
 - Precise positioning
 - Longer operating life
-

74. FUTURE MOTOR CONTROLLER

Current:

MDD10A-class

Future:

- Industrial motor controller
- CAN motor controller
- BLDC controller
- Servo controller
- Regenerative braking where appropriate

Architecture:

ROS 2

↓

Robot controller

↓

CAN

↓

Industrial motor controller

↓

BLDC/Servo

75. FUTURE CUSTOM PCB

Current V1:

Modular wiring

Development boards

Individual modules

Future:

Custom KiCad/Altium PCB

Integration:

ESP32

Motor interface

IMU

Encoder interfaces

Ultrasonic

Bumpers

E-stop monitoring

Power monitoring

CAN/UART

Debug headers

This is specifically identified as an industry-grade upgrade in the uploaded architecture.

76. FUTURE POWER SYSTEM

Current:

Dual LiPo
+
Regulators

Future:

Smart BMS
+
Protected battery
+
Fuel gauge
+
Current sensing
+
Voltage monitoring
+
Thermal monitoring
+
Protected PDB

Additional protections:

- Reverse polarity
- Over-current
- Under-voltage
- Battery health telemetry

The source architecture explicitly lists these future power-system upgrades.

77. FUTURE CHASSIS

Current:

DIY acrylic/aluminium

Future:

CNC aluminium
+
Industrial polymer panels
+
Sealed electronics compartment
+
Serviceable battery tray
+
Proper cable harness

This allows:

- Better rigidity
 - Better serviceability
 - Better protection
 - Better repeatability
-

78. FUTURE NAVIGATION

Current:

RPLiDAR
+
Encoder
+
IMU
+
Ultrasonic
+
Nav2

Future:

Premium 2D LiDAR
+
3D LiDAR
+
Depth camera
+
VIO

+
Advanced localization
+
Multi-floor navigation
+
Elevator integration

The source overview defines this exact general upgrade direction.

79. FUTURE VISION

Current:

USB Camera
+
OpenCV

Future:

- Depth camera
 - Stereo camera
 - Better low-light RGB camera
 - Hardware-accelerated vision
 - People detection
 - Semantic navigation
 - Object recognition
-

80. FUTURE NETWORKING

Current:

Wi-Fi
BLE
Ethernet

Future:

Industrial Ethernet

+

CAN

+

Secured MQTT

+

VLAN segmentation

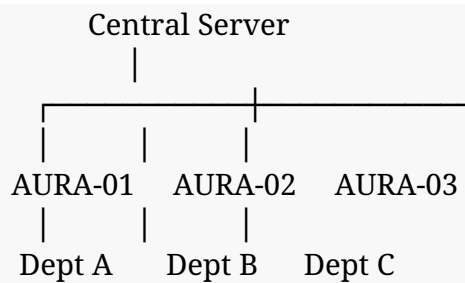
+

Enterprise authentication

This becomes important when multiple robots are deployed.

81. FUTURE FLEET ARCHITECTURE

Instead of one robot:



Each robot reports:

- Robot ID
- Location
- Battery
- Health
- Firmware version
- Logs
- Mission state

This fleet direction is explicitly included in the source architecture.

82. OTA UPDATE SYSTEM

Future industrial architecture:

Central Cloud



Signed Release



Robot



OTA Update



Verification



Success

If failure:

Failed update



Rollback



Previous stable version

This should eventually apply to:

- ESP32 firmware
 - ROCK software
 - AI models
 - Navigation maps
 - Configuration
-

83. SECURITY EVOLUTION

V1

Authentication
RBAC
Device binding
BLE
TOTP

Industrial version

TLS
+
Certificate-based robot identity
+
Secure boot
+
Encrypted secrets
+
Signed firmware
+
Key rotation
+
Audit logs
+
Role separation
+
Network segmentation

The source architecture explicitly distinguishes these V1 and industrial security levels.

84. SAFETY EVOLUTION

V1

Physical E-stop
+
Bumpers
+
Ultrasonic

+
LiDAR
+
Software watchdog

Industrial

Physical E-stop
+
Dual-channel safety
+
Motor power contactor
+
Safety-rated sensors
+
Hardware watchdog
+
Fault logging
+
Safe-state recovery
+
Certification-oriented design

The uploaded overview defines this progression.

85. RESEARCH VALUE OF AURA

AURA can support several genuine engineering research areas.

1. Edge AI for robotics

Compare:

Local AI
vs
Cloud AI

Measure:

- Latency
- Accuracy

- Reliability
 - Resource usage
-

2. Hybrid navigation

Compare:

LiDAR only

against:

LiDAR

+

Encoder

+

IMU

+

ArUco

Measure localization improvement.

3. Department-specific RAG

Compare:

General LLM

vs

Department RAG + LLM

Measure:

- Factual accuracy
 - Hallucination
 - Response latency
 - Domain adherence
-

4. Multimodal educational AI

Possible inputs:

PDF

+

Handwritten notes

+

Circuit diagrams

+

CRO/DSO waveforms

5. Human-robot interaction

Compare:

Voice

+

Touchscreen

+

Mobile

+

Physical navigation

as one integrated HRI system.

These research directions are already present in the detailed source overview.

86. WHAT TO MEASURE

Do not only say:

"The robot works."

Measure it.

Navigation

- Position error
- Route completion rate

- Obstacle avoidance success
- Stopping error
- Navigation time

Voice

- Word error rate
- Command recognition rate
- Response latency

RAG

- Retrieval accuracy
- Answer correctness
- Hallucination rate

Robot

- Battery consumption
- Motor temperature
- ROCK temperature
- CPU usage
- RAM usage
- Uptime

Attendance

- False acceptance
- False rejection
- QR validation time
- BLE reliability

These measurements are defined in the source architecture.

87. V1 PERFORMANCE TARGETS

These should be treated as **engineering targets**, not claims that the robot already achieves them.

Metric	Target
Route completion	$\geq 95\%$
Safety stop	100% in controlled tests
Attendance validation	$\geq 99\%$
Basic department query correctness	$\geq 95\%$
Voice command recognition	$\geq 90\%$
Local deterministic command response	<1 s target
Normal navigation speed	0.15–0.25 m/s
Daily operation	5–6 trips
General voice interaction	~5 s practical target

The source overview contains the same general performance-target framework.

88. CURRENT V1 LIMITATIONS

AURA V1 must be reported honestly.

Hardware

- Budget motors
- Prototype chassis
- Consumer LiDAR
- Consumer sensors
- Non-industrial battery system

AI

- Small local model
- Cloud dependency for some multimodal functions
- Model performance depends on runtime/quantization support

Navigation

- Indoor
- Single floor
- Mapped environment
- Controlled speed

Attendance

- BLE RSSI is not exact distance
- Device binding depends on mobile platform
- QR security depends on backend validation

Faculty recognition

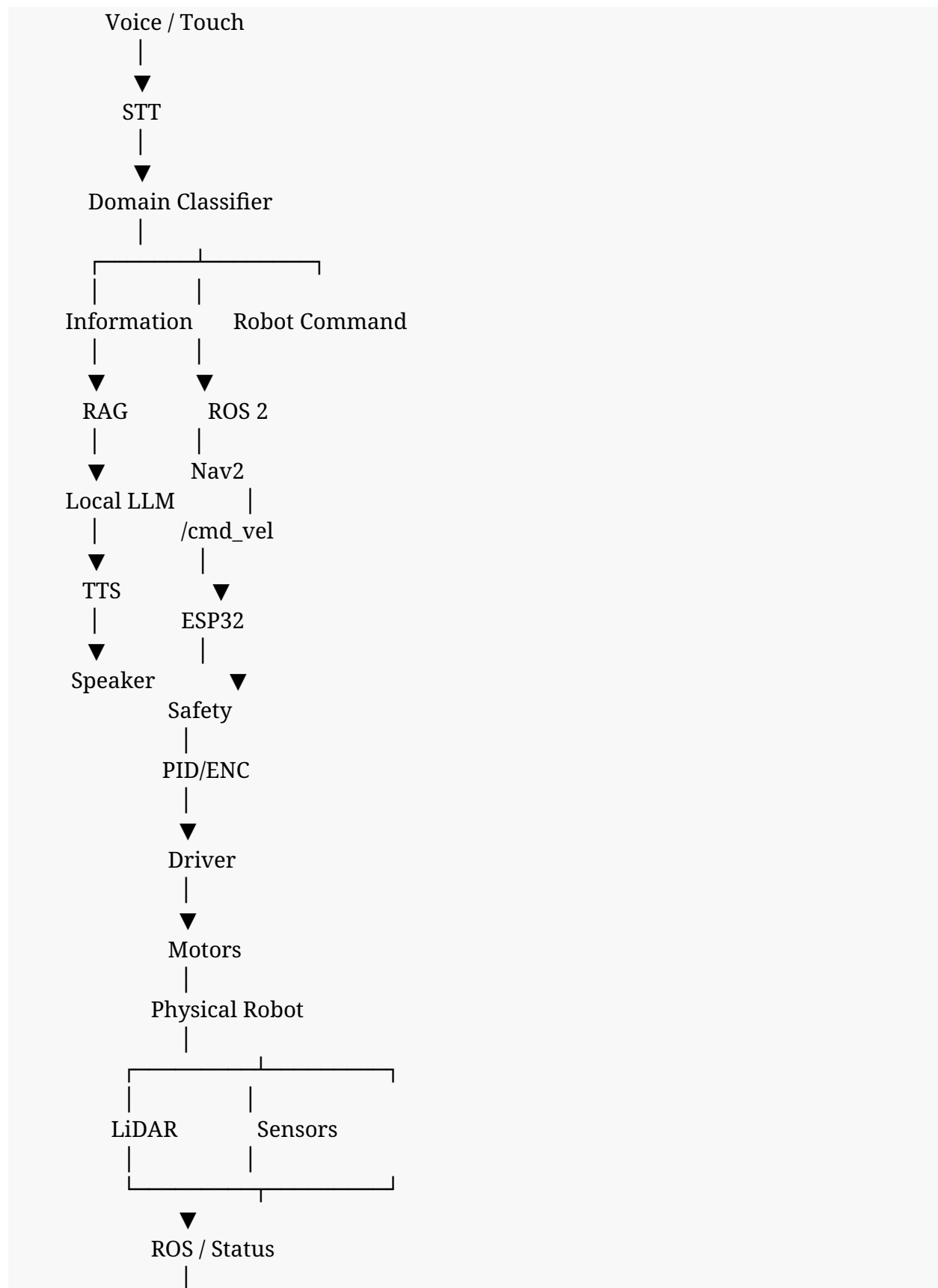
- Small registered faculty set
- Lighting affects recognition
- Camera angle affects recognition

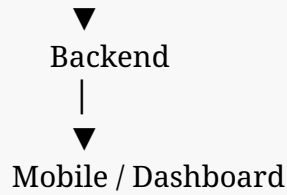
These limitations are consistent with the uploaded overview.

89. THE MOST IMPORTANT SYSTEM INTEGRATION

The entire AURA project can ultimately be demonstrated through one chain:







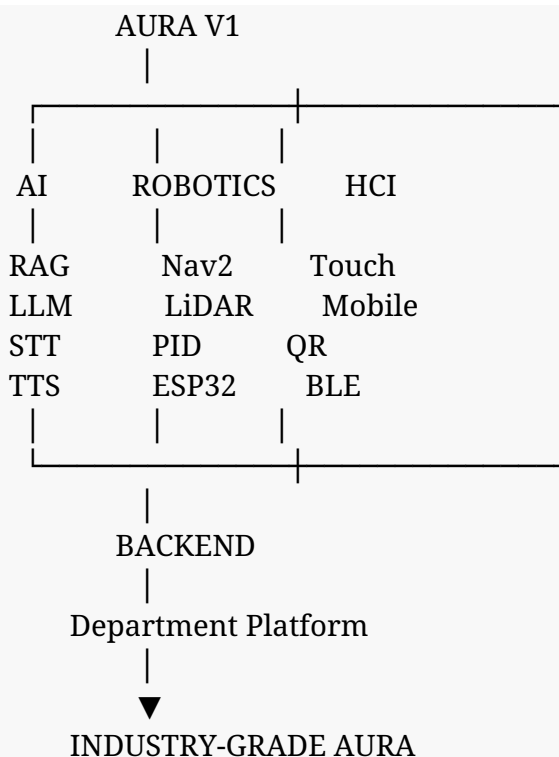
This is the **actual engineering story of AURA**.

90. FINAL DEFINITION OF THE NEW V1

AURA V1 is:

A department-specific, voice-first, edge-AI autonomous assistant robot combining local RAG/AI, ROS 2/Nav2 navigation, real-time ESP32 motor control, LiDAR/encoder/IMU-based navigation, controlled faculty recognition, touchscreen/mobile interaction, secure QR/BLE attendance, and department backend services, designed as a ₹45K prototype that can evolve into an industry-grade indoor service and educational platform.

The fundamental philosophy is:



The key engineering rule

Do not try to make V1 industrial-grade by adding expensive hardware everywhere.

Instead:

₹45K hardware



Correct architecture



Measurement



Calibration



Closed-loop control



Simulation



Testing



Reliable V1



Upgrade-ready foundation

That is the correct way to turn the new V1 into a serious engineering platform rather than just an expensive demonstration robot. The uploaded overview itself emphasizes that V1 does not need industrial-grade hardware to provide an industrial-grade engineering foundation.